

Avery Doiron

CONTACT

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EDUCATION

St. Francis Xavier University

Antigonish, NS, Sep. 2023 – Apr. 2027 (Expected)

Bachelor of Science, Honours in Computer Science, Minor in Mathematics, Computing Concentration

GPA: 3.9/4.0 | Major GPA: 4.0/4.0

Honours Thesis

Applying evolutionary computation to evolve strategies for classic game-theoretic games under the supervision of Dr. James Hughes (in progress).

Relevant Coursework

Sports Analytics

Data Science

Statistical Methods

Artificial Intelligence

Evolutionary Computation

Natural Language Processing

Database Management Systems

Software Design

Algorithm Design and Analysis

Awards and Scholarships

Dean's List

2025 & 2026

In-Course Scholarship (\$1,000 CAD)

2024, 2025, & 2026

Alley Heaps Undergraduate Research Award (\$9,250 CAD)

2026

EXPERIENCE

Hockey Operations Analyst Volunteer

Owen Sound Attack, Ontario Hockey League (OHL)

Remote, Sep. 2026 – Present

- Support the collection, organization, and analysis of hockey operations data for the organization's newly formed analytics and research department
- Conduct video analysis and game tagging to help identify player and team performance trends throughout the OHL season
- Build reports and visualizations translating raw data into insights for coaching staff
- Support research projects related to player evaluation and development
- Collaborate with team staff on department initiatives

Undergraduate Research Intern

St. Francis Xavier University

Antigonish, NS, Apr. 2026 – Aug. 2026

- Conducted research on the relationship between NHL personnel interview language and on-ice performance under the supervision of Dr. Milton King and Dr. Kyran Cupido
- Built a data pipeline to scrape, clean, and label 5,000+ interview transcripts, linking each to speaker identity, interview timing, game results, and player statistics
- Applied transformer-based sentiment and emotion models to quantify linguistic features
- Evaluated relationships between linguistic features and performance using statistical testing
- Visualized sentiment trends, embedding clusters, and outlier interviews to communicate findings

Junior Technical Analyst

SEASI

Remote, May. 2024 – Jun. 2024 & May. 2025 – Aug. 2025

- Managed Microsoft Dynamics CRM data for 100+ consultant profiles and built dashboards that improved reporting workflows
- Designed backend database infrastructure on Microsoft Azure, earning the AZ-900 certification
- Contributed to backend system design and implementation for the company website and an AI chatbot
- Customized 50+ consultant proposals to align with federal RFP requirements, contributing to multiple successful contract awards
- Supported testing, documentation, and implementation of cloud-based IT tools

PROJECTS

NHL Player and Team Evaluation Cards

Jan. 2025 – Present

- Built an automated pipeline to scrape, clean, and structure multi-season NHL data
- Trained an expected goals (xG) model to quantify shot quality (0.779 AUC on 1.3M+ shots)
- Built a regularized adjusted plus-minus (RAPM) model to isolate individual player impact from teammates and opponents
- Converted player impact into Wins Above Replacement (WAR) using a custom goals-to-wins conversion
- Automated generation of 12,300+ player cards and 500+ team cards across 17 seasons

NHL Schedule Optimization Genetic Algorithm

Sep. 2025 – Dec. 2025

- Designed a genetic algorithm in Python to generate optimized NHL schedules under real-world league constraints
- Built a multi-objective fitness function balancing rest, streak fairness, and travel distance across a full 1,344-game season
- Built custom genetic operators and adaptive selection pressure, improving fitness ~2,000x over 10,000 generations
- Validated configuration choices with an 800-run hyperparameter sensitivity study
- Built visualizations and presented findings in a 40-minute talk

TECHNICAL SKILLS

Programming Languages

Python, R, SQL, Java, JavaScript, HTML, CSS

Libraries/Frameworks

Pandas, NumPy, SciPy, Scikit-learn, SQLite, BeautifulSoup, Matplotlib, Plotly, PyTorch, Hugging Face

Tools & Platforms

Git/GitHub, Visual Studio Code, PyCharm, RStudio, Jupyter Notebook, Microsoft Office